Integrals

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1 Derivative Formulas for Revision

$$\frac{d}{dx}x^n = nx^{n-1} \qquad \frac{d}{dx}e^x = e^x$$

$$\frac{d}{dx}\ln|x| = \frac{1}{x} \qquad \frac{d}{dx}a^x = a^x \ln a$$

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\cos x = -\sin x$$

$$\frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx}\sec x = \sec x \tan x \qquad \frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2} \qquad \frac{d}{dx}\cot^{-1} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}} \qquad \frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}.$$

For inverse secant and inverse cosecant, the standard real derivative expressions use $|x|\sqrt{x^2-1}$ in the denominator.

2 Basic Indefinite Integrals

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int 1 \, dx = x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x \, dx = -\cot x + C$$

$$\int \frac{dx}{x} = \ln |x| + C$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C.$$

3 Partial Fractions

For distinct linear factors,

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}.$$

For a repeated linear factor,

$$\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}.$$

For three distinct linear factors,

$$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}.$$

For a quadratic irreducible factor,

$$\frac{p(x)}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

when the numerator has sufficiently low degree.

4 Special Trigonometric Integrals

$$\int \tan x\,dx = \ln|\sec x| + C$$

$$\int \cot x\,dx = \ln|\sin x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C$$

$$\int \operatorname{cosec} x\,dx = \ln|\operatorname{cosec} x - \cot x| + C.$$

5 Standard Special Forms

For $a > 0$,

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln|x + \sqrt{x^2 - a^2}| + C$$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln|x + \sqrt{x^2 + a^2}| + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C.$$

6 Integration by Substitution and Parts

For a substitution $u = g(x)$, rewrite $du = g'(x)\,dx$ and integrate in u.

$$\int e^x\big(f(x) + f'(x)\big)\,dx = e^x f(x) + C.$$

Integration by parts:

$$\boxed{\int u\,dv = uv - \int v\,du.}$$

The notes suggest choosing u using the ILATE order: inverse trigonometric, logarithmic, algebraic, trigonometric, exponential.

7 Definite Integrals

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad F'(x) = f(x).$$

7.1 Linearity

$$\int_a^b (f+g)\,dx = \int_a^b f\,dx + \int_a^b g\,dx$$
$$\int_a^b (f-g)\,dx = \int_a^b f\,dx - \int_a^b g\,dx$$
$$\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx.$$

7.2 Definite Integral Properties

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$
$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$
$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$
$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx.$$

7.3 Symmetry

$$\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & f \text{ even}, \\ 0, & f \text{ odd}. \end{cases}$$
$$\int_0^{2a} f(x)\,dx = \int_0^a \big(f(x) + f(2a-x)\big)\,dx.$$

8 Square Root Integrals

For $a > 0$,

$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln|x+\sqrt{x^2-a^2}| + C$$
$$\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln|x+\sqrt{x^2+a^2}| + C$$
$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C.$$